



Treewidth and minimum fill-in on permutation graphs in linear time[☆]

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ABSTRACT

Permutation graphs form a well-studied subclass of cocomparability graphs. Permutation graphs are the cocomparability graphs whose complements are also cocomparability graphs. A triangulation of a graph G is a graph H that is obtained by adding edges to G to make it chordal. If no triangulation of G is a proper subgraph of H then H is called a minimal triangulation. The main theoretical result of the paper is a characterisation of the minimal triangulations of a permutation graph, that also leads to a succinct and linear-time computable representation of the set of minimal triangulations. We apply this representation to devise linear-time algorithms for various minimal triangulation problems on permutation graphs, in particular, we give linear-time algorithms for computing treewidth and minimum fill-in on permutation graphs.

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1. Introduction

Treewidth and minimum fill-in are among the most interesting graph parameters. The treewidth of a graph is a measure of its treelikeness, and the minimum fill-in can be understood as a measure of the degree of chordality of a graph. Both parameters can be formulated as embedding problems into chordal graphs where, in the case of treewidth, the clique number and in the case of minimum fill-in, the number of edges, are to be minimized. Treewidth plays a great role in algorithm design, since many hard problems can be solved efficiently on graphs of bounded treewidth. Minimum fill-in has applications in Gaussian elimination of symmetric sparse matrices [16]. The corresponding decision problems TREewidth and MINIMUM FILL-IN are NP-complete even on cobipartite graphs [1,18].

A triangulation of a graph G is a chordal graph H where G and H have the same vertex set and every edge of G is an edge of H . If no edge of H can be deleted so that the resulting graph still forms a triangulation of G then H is called a *minimal triangulation* of G . Minimal triangulations of graphs were first studied by Rose et al. [17]. Using the notion of minimal triangulation, treewidth and minimum fill-in admit the following definition. For an arbitrary graph G , the following holds:

- let H be a minimal triangulation of G of smallest clique number among the minimal triangulations of G ; the *treewidth* of G , denoted as $\text{tw}(G)$, is equal to the clique number of H minus 1
- let H be a minimal triangulation of G with smallest number of edges among the minimal triangulations of G ; the *minimum fill-in* of G , denoted as $\text{mfi}(G)$, is equal to the number of edges of H minus the number of edges of G .

A clique of a graph is a set of vertices that are pairwise adjacent, and the clique number of a graph is the maximum number of vertices in a clique. Motivated also by the above connection, much work has been devoted to studying and understanding minimal triangulations of general as well as special graphs. Within the focus of this paper, interesting results are: minimal triangulations of cographs are cographs [4], and minimal triangulations of permutation graphs, cocomparability graphs

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and AT-free graphs are interval graphs [2,10,13]. The inclusion structure of the mentioned graph classes is: cographs $\subset$ permutation graphs $\subset$ comparability graphs $\subset$ AT-free graphs. For general graphs, Parra and Scheffler as well as Kloks, Kratsch, Spinrad characterised minimal triangulations using minimal separators [14,11].

In this paper, we will give linear-time algorithms for computing treewidth and minimum fill-in on permutation graphs. In particular, the running times of our algorithms will be independent of the computed parameter. The previously best known algorithms for the two problems are due to the work of Bodlaender, Kloks, Kratsch and Bodlaender, Kloks, Kratsch, Müller, respectively, with the respective running times $\mathcal{O}(\text{tw}(G) \cdot n)$ and $\mathcal{O}(n^2)$ [2,3]. The algorithms in this paper are based on a succinct representation of the minimal triangulations of a permutation graph. This characterisation is the main theoretical result of the paper. Using this representation, treewidth and minimum fill-in become shortest-path problems in a weighted acyclic directed graph. A similar approach was applied by Bodlaender, Kloks, Kratsch for their treewidth algorithm [2].

The characterisation of the minimal triangulations of a permutation graph that we present in this paper partitions vertices into groups. Each group forms a maximal clique in a minimal triangulation of the graph. Such sets of vertices, that form maximal cliques in a minimal triangulation, are called *potential maximal cliques*. The notion of potential maximal clique was introduced by Bouchitté and Todinca [5]. They showed for instance that treewidth and minimum fill-in are polynomial-time solvable on graphs with a polynomial number of potential maximal cliques [5,6]. Permutation graphs have polynomially many potential maximal cliques. In fact, it follows from our results that the number of potential maximal cliques of permutation graphs is linear in the number of vertices and edges of the graph. Note that this does not imply a polynomial bound on the number of minimal triangulations, since permutation graphs can have an exponential number of minimal triangulations. In addition to the idea of grouping vertices into potential maximal cliques, our characterisation relies on the fact that every minimal triangulation of a permutation graph is an interval graph [2].

The paper is organized as follows. In Section 2, we define the necessary graph notions and give definitions and properties of minimal triangulations and of permutations and interval graphs. In Section 3, we introduce an operation on permutation graphs that produces minimal triangulations. The main theoretical result of the paper is presented in Section 4. We show that the defined operation can produce **every** minimal triangulation of a permutation graph. Linear-time algorithms for treewidth and minimum fill-in and further problems on permutation graphs are given in the last section.

2. Definitions and notation

We consider only simple and finite graphs, directed and undirected. If not said otherwise, when we speak of “graph” we always mean undirected graph. Directed graphs, *digraphs* for short, appear only as auxiliary structures. For a graph $G = (V, E)$, the vertex set of G is $V(G) = V$ and the edge set of G is $E(G) = E$. An edge of G is denoted as uv . For a vertex pair u, v of G , if uv is an edge of G , we say that vertices u and v are *adjacent*. A graph G' is a *subgraph* of G , denoted as $G' \subseteq G$, if $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$. For $S \subseteq V(G)$, the subgraph of G induced by S , denoted as $G[S]$, is the subgraph of G on vertex set S and all edges uv of G with $u, v \in S$. We write $G \setminus S$ instead of $G[V \setminus S]$. For F a set of edges, $G \cup F$ denotes the graph $(V(G), E(G) \cup F)$. The graph $(V(G), E(G) \setminus \{e\})$ where $e \in E(G)$ is denoted as $G - e$, and $G \setminus F =_{\text{def}} (V(G), E(G) \setminus F)$. The *(open) neighbourhood* of a vertex u , denoted as $N_G(u)$, is the set of all vertices adjacent to u in G ; $N_G[u] =_{\text{def}} N_G(u) \cup \{u\}$ denotes the *closed neighbourhood* of u . A *clique* of G is a set of vertices that are pairwise adjacent. The *clique number* of G is the largest number of vertices in a clique of G . A clique is *maximal* if it is not properly contained in another clique. For G' , a digraph, we denote the arcs of G' as (u, v) . A sequence $(u_1, \dots, u_k)$ of pairwise different vertices of G' where $k \geq 2$ is called a *cycle* if $(u_1, u_2), \dots, (u_{k-1}, u_k), (u_k, u_1)$ are arcs of G . A digraph without a cycle is called *acyclic*. Digraphs and particularly acyclic digraphs will only appear as auxiliary graphs in Section 5.

Let G be a graph. For vertices u, v of G , a u, v -path of length k of G is a sequence $(u_0, \dots, u_k)$ of pairwise different vertices of G where $u_0 u_1, \dots, u_{k-1} u_k$ are edges of G and $u_0 = u$ and $u_k = v$. A graph is *connected* if it has a u, v -path for every vertex pair u, v . A *connected component* of G is a connected induced subgraph $G[S]$ that is not properly contained in another connected subgraph of G . Let $S \subseteq V(G)$. We call S an a, b -separator of G for a pair a, b of non-adjacent vertices of G if a and b are vertices in different connected components of $G \setminus S$. S is a *minimal a, b -separator* of G if there is no a, b -separator S' with $S' \subset S$. S is a *minimal separator* of G if S is a minimal a, b -separator for some non-adjacent vertices $a, b \in V$. A connected component C of $G \setminus S$ is S -full if every vertex of S has a neighbour in C ; S -full components of G are S -full connected components of $G \setminus S$.

Lemma 2.1 (Folklore). *Let G be a graph and let a and b be vertices of G . A set S of vertices of G is a minimal a, b -separator of G if and only if G has two (different) S -full components where one contains a and the other contains b .*

Let G be a graph. A cycle of length k of G is a path $(u_1, \dots, u_k)$ with $u_1 u_k$ an edge of G . A *chord* in a cycle $(u_1, \dots, u_k)$ is an edge $u_i u_j$ where $1 < i + 1 < j \leq k$ and $u_i u_j \neq u_1 u_k$. Informally, a chord joins two non-consecutive vertices of the cycle. A graph that has a chord in every cycle of length at least 4 is called *chordal*. In other words, a chordal graph contains no cycle of length at least 4 as induced subgraph. Let H be a graph on vertex set $V(G)$. If $G \subseteq H$ and H is chordal, we call H a *triangulation* of G . If there is no triangulation H' of G with $G \subseteq H' \subset H$, we call H a *minimal triangulation* of G . Minimal triangulations have a nice characterisation.

Proposition 2.2 ([17]). *Let G and $H = G \cup F$ be graphs where H is chordal and $E(G) \cap F = \emptyset$. Then, H is a minimal triangulation of G if and only if for every $e \in F$, e is the unique chord in a cycle of length 4 in H .*

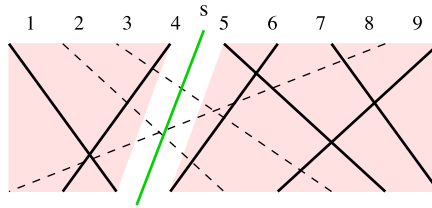


Fig. 1. Scanline s partitions the vertex set of the permutation graph into three sets: left vertices and right vertices (full line segments) and crossed vertices (dashed line segments).

In this article, we mainly consider two classes of graphs, namely permutation graphs and interval graphs. For $n \geq 1$, a *permutation* over $\{1, \dots, n\}$ is a bijective function from $\{1, \dots, n\}$ to $\{1, \dots, n\}$. Every permutation defines a graph. Let π be a permutation over $\{1, \dots, n\}$. The graph defined by π , denoted as $G(\pi)$, has vertex set $\{1, \dots, n\}$ and two vertices u, v are adjacent if and only if $(u - v) \cdot (\pi^{-1}(u) - \pi^{-1}(v)) < 0$. For $u, v \in \{1, \dots, n\}$, we write $u \prec_{\pi} v$ if $\pi^{-1}(u) \leq \pi^{-1}(v)$. If $u \prec_{\pi} v$ and $u \neq v$, we write $u \prec_{\pi} v$. Then, two vertices u, v of $G(\pi)$, where $u < v$, are adjacent if and only if $v \prec_{\pi} u$. A graph G on vertex set $\{1, \dots, n\}$ is called a *permutation graph* if there is a permutation π over $\{1, \dots, n\}$ with $G = G(\pi)$. Note that permutation graphs are often defined as being *isomorphic* to some $G(\pi)$. The more restrictive definition that we use here is more convenient in proofs. Permutation graphs have a nice geometric model in the plane, called *permutation diagram*: every vertex has a line segment, line segments have their endpoints on two horizontal lines, and two vertices are adjacent if and only if the corresponding line segments have a non-empty intersection. More information about permutation graphs and permutation diagrams can be found in the book by Golumbic [9]. Our definitions and proofs will not rely on permutation diagrams, but they provide a good tool for understanding.

The second used graph class is interval graphs. A graph whose vertices can be assigned closed intervals of the real line such that two vertices are adjacent if and only if the corresponding assigned intervals have a non-empty intersection is called *interval graph*. Interval graphs are chordal. For a graph G and sets $S_1, \dots, S_k$ of vertices of G , the linear arrangement $\langle S_1, \dots, S_k \rangle$ has the *consecutive ones property* if for every vertex x of G and every i, j with $1 \leq i < j \leq k$, $x \in S_i \cap S_j$ implies $x \in S_i \cap \dots \cap S_j$. A *consecutive clique arrangement* for a graph G is a linear arrangement $\langle A_1, \dots, A_k \rangle$ of its maximal cliques that has the consecutive ones property. A graph is an interval graph if and only if it has a consecutive clique arrangement. Interval graphs play an important role for permutation graphs, since every minimal triangulation of a permutation graph is an interval graph [2].

3. Minimal triangulations from permutation diagrams

We will give a characterisation of the minimal triangulations of a permutation graph. This characterisation is based on a partition operation over the set of endpoints of the line segments in a permutation diagram. The main result in this section shows that this operation always produces a minimal triangulation. The partition operation is defined by use of the so-called *scanlines*. Scanlines were introduced by Bodlaender et al. [2].

Definition 3.1. Let $G = G(\pi)$ be a permutation graph on n vertices. A *scanline* of G is a pair (p, q) of numbers where $p, q \in \{\frac{1}{2}, 1\frac{1}{2}, \dots, n\frac{1}{2}\}$. The *enclosure* of scanline (p, q) is formed by the vertices $a = p - \frac{1}{2}, b = p + \frac{1}{2}, c = \pi(q - \frac{1}{2}), d = \pi(q + \frac{1}{2})$ and denoted as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

As a convention, in this paper, we assume that a permutation graph on n vertices has the two non-visible vertices 0 and $n + 1$ and that $\pi(0) = 0$ and $\pi(n + 1) = n + 1$. This is a technical assumption that makes the definition of the enclosure of a scanline easier.

The scanlines $(\frac{1}{2}, \frac{1}{2})$ and $(n\frac{1}{2}, n\frac{1}{2})$ play a special role; they are denoted by the symbols **0** and **1**, respectively. Let $G = G(\pi)$ be a permutation graph on n vertices and let $s = (p, q)$ and $s' = (p', q')$ be (different) scanlines of G . We say that s is *to the left of* s' if $p \leq p'$ and $q \leq q'$, denoted as $s < s'$. If $s < s'$ or $s' < s$, we say that s and s' are *non-intersecting* or *do not intersect*; otherwise they *intersect*. Note that for every scanline s of G with $s \neq \mathbf{0}, \mathbf{1}$, it holds that $\mathbf{0} < s < \mathbf{1}$. Every scanline partitions the vertex set of G into three sets: the vertices to the left, the vertices to the right and the vertices that are crossed. Formally, for $s = (p, q)$ a scanline of G , these sets are defined as follows:

- $L(s) =_{\text{def}} \{x \in V(G) : x < p \text{ and } \pi^{-1}(x) < q\}$
- $R(s) =_{\text{def}} \{x \in V(G) : p < x \text{ and } q < \pi^{-1}(x)\}$
- $C(s) =_{\text{def}} V(G) \setminus (L(s) \cup R(s))$.

We say that the vertices in $L(s)$ are *to the left of* s , the vertices in $R(s)$ are *to the right of* s , and the vertices in $C(s)$ are *crossed by* s . An example is given in Fig. 1: the vertices to the left and right of scanline s are entirely contained in the two filled regions.

The enclosure of s is $\begin{pmatrix} 4 & 5 \\ 1 & 6 \end{pmatrix}$.

Let $\mathcal{S} = \{s_1, \dots, s_k\}$ be a set of pairwise non-intersecting scanlines of G where $s_1 < \dots < s_k$. The scanlines in $\mathcal{S}$ partition the endpoints of the line segments in the permutation diagram of G . This partition can define an interval graph. For every vertex x of G , we define two numbers:

$$l(x) =_{\text{def}} \min\{i : x \in L(s_i) \cup C(s_i)\} \cup \{k+1\}$$

$$r(x) =_{\text{def}} \min\{i : x \in L(s_i)\} \cup \{k+1\}.$$

It is clear from the definition that $l(x) \leq r(x)$ for every vertex x of G . We call the function pair (l, r) the *endpoint function pair* of $\mathcal{S}$ in G . When the context G and $\mathcal{S}$ is clear, we will also refer to the pair as “endpoint function pair”. By $\mathcal{F}_G(\mathcal{S})$, we denote the family $\{[l(x), r(x)]\}_{x \in V(G)}$ of closed intervals of the real line. And by $G(\mathcal{S})$, we denote the interval graph that is defined by $\mathcal{F}_G(\mathcal{S})$.

Lemma 3.2. *Let $G = G(\pi)$ be a permutation graph and let $\mathcal{S}$ be a set of pairwise non-intersecting scanlines of G .*

- (1) $G(\mathcal{S})$ is a triangulation of G .
- (2) For every $\mathcal{S}' \subseteq \mathcal{S}$, $G(\mathcal{S}')$ is a subgraph of $G(\mathcal{S})$.

Proof. Let (l, r) be the endpoint function pair of $\mathcal{S}$ in G and let $k =_{\text{def}} |\mathcal{S}|$. We show the first statement. Since $G(\mathcal{S})$ is an interval graph and thus chordal, it remains to show that G is a subgraph of $G(\mathcal{S})$. Let u and v be a pair of non-adjacent vertices of $G(\mathcal{S})$. Without loss of generality, we can assume that $r(u) < l(v)$. Since $r(u) \leq k$, there is a scanline $s = (p, q)$ in $\mathcal{S}$ with $u \in L(s)$ and $v \notin L(s) \cup C(s)$. The latter condition is equivalent to $v \in R(s)$. Thus, $u < v$ and $u \prec_{\pi} v$, which means that u and v are non-adjacent in G .

For the second statement, we show that every pair of non-adjacent vertices of $G(\mathcal{S}')$ is non-adjacent in $G(\mathcal{S})$. Let (l', r') be the endpoint function pair of $\mathcal{S}'$. Let u and v be a pair of non-adjacent vertices of $G(\mathcal{S}')$. Without loss of generality, we can assume that $r'(u) < l'(v)$. By similar arguments as above, there is a scanline s in $\mathcal{S}'$ that separates u and v . Since $s \in \mathcal{S}$, there is a scanline in $\mathcal{S}$ that separates u and v in the same sense, and $r(u) < l(v)$ follows. Hence, u and v are non-adjacent in $G(\mathcal{S})$. $\square$

It is not difficult to see that there are triangulations of a permutation graph that cannot be obtained from our construction. A simple example is the path on five vertices: a triangulation where the two endpoints are adjacent, makes the middle vertex adjacent to the two endpoints. However, there is a triangulation of the path on five vertices with the two endpoints adjacent but none of the endpoints are adjacent to the middle vertex.

We are interested not in arbitrary triangulations but in minimal triangulations. The main result of this section gives a sufficient condition on the set of scanlines to produce a minimal triangulation. This condition involves a special class of scanlines.

Definition 3.3. Let $G = G(\pi)$ be a permutation graph. A scanline of G is called *minimal separation line* if it does not cross any of its enclosure vertices.

Every permutation graph has minimal separation lines, for example **0** and **1**. For a complete permutation graph, these are the only minimal separation lines. For non-complete permutation graphs, more minimal separation lines exist, as can be seen by the following procedure in a permutation diagram: choose a pair of non-adjacent vertices, starting from the left vertex, find the first non-neighbour to the right on the upper and lower lines. These two vertex appearances are part of the enclosure of a minimal separation line. This also shows that for every pair of non-adjacent vertices, there is a minimal separation line that separates the two in the sense of the proof of Lemma 3.2. More can be said about minimal separation lines. In fact, a result by Bodlaender et al. in [2] can be extended to the following statement and proved similarly.

Lemma 3.4. *Let $G = G(\pi)$ be a permutation graph. Let X and Y be sets of vertices of G such that $G[X]$ and $G[Y]$ are connected. If there is no pair x, y with $x \in X$ and $y \in Y$ and $xy \in E(G)$ then there is a minimal separation line s of G such that $X \subseteq L(s)$ and $Y \subseteq R(s)$ or $Y \subseteq L(s)$ and $X \subseteq R(s)$ and $C(s)$ is a minimal x, y -separator of G for some vertices $x \in X$ and $y \in Y$.*

Let $G = G(\pi)$ be a permutation graph. Let s and t be minimal separation lines of G . We call t a *successor* of s if $s < t$ and there is no minimal separation line s' of G such that $s < s' < t$. Note that every minimal separation line of G except for **1** has a successor.

Lemma 3.5. *Let $G = G(\pi)$ be a permutation graph. Let s and t be minimal separation lines of G where t is a successor of s . Then, the following holds:*

- (1) $R(s) \cap L(t)$ is a clique in G
- (2) for every $x \in R(s) \cap L(t)$ and $y \in C(s) \cup C(t)$, $xy \in E(G)$
- (3) for every $x \in C(s) \cap L(t)$ and $y \in R(s) \cap C(t)$, $xy \in E(G)$.

Proof. Suppose that there is a pair x, y of non-adjacent vertices of G with $x < y$ and $x \in (C(s) \setminus C(t)) \cup (R(s) \cap L(t))$ and $y \in (C(t) \setminus C(s)) \cup (R(s) \cap L(t))$. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ be the enclosures of respectively s and t . We distinguish between four cases with respect to x and y and obtain the following results by the application of Lemma 3.4:

case	minimal separation line s' exists
$x \in C(s)$ and $y \in C(t)$	$a, c, x \in L(s')$ and $b', d', y \in R(s')$
$x \notin C(s)$ and $y \in C(t)$	$x \in L(s')$ and $b', d', y \in R(s')$
$x \in C(s)$ and $y \notin C(t)$	$a, c, x \in L(s')$ and $y \in R(s')$
$x \notin C(s)$ and $y \notin C(t)$	$x \in L(s')$ and $y \in R(s')$

which shows that $a, c, x \in L(s')$ and $b', d', y \in R(s')$ in all the four cases. Hence, $s < s' < t$, which is a contradiction to t being successor of s . The lemma follows by observing that $C(s) \cap L(t) = C(s) \setminus C(t)$ and $R(s) \cap C(t) = C(t) \setminus C(s)$ and that every vertex in $C(s) \cap C(t)$ is adjacent to every vertex in $R(s) \cap L(t)$ by the properties of permutation diagrams. $\square$

Theorem 3.6. Let $G = G(\pi)$ be a permutation graph and let $\mathcal{T}$ be a set of pairwise non-intersecting scanlines. If $\mathcal{T}$ contains a maximal set of pairwise non-intersecting minimal separation lines then $G(\mathcal{T})$ is a minimal triangulation of G .

Proof. Let $\mathcal{S}$ be the set of minimal separation lines in $\mathcal{T}$. We assume that $\mathcal{S}$ is a maximal set of pairwise non-intersecting minimal separation lines of G . We first show that $G(\mathcal{S})$ is a minimal triangulation of G . Due to Lemma 3.2, $G(\mathcal{S})$ is a triangulation of G . For proving minimality, we apply Proposition 2.2 and show that every edge in $E(G(\mathcal{S})) \setminus E(G)$ is the unique chord in a cycle of length 4 in $G(\mathcal{S})$. Let (l, r) be the endpoint function pair of $\mathcal{S}$. Let u, v be a pair of adjacent vertices of $G(\mathcal{S})$. Without loss of generality, we can assume that $l(u) \leq l(v)$. Since $r(u) < l(v)$ means that u and v are non-adjacent in $G(\mathcal{S})$, it holds that $l(u) \leq l(v) \leq r(u)$. If $l(u) < l(v) \leq r(v) < r(u)$ then there are minimal separation lines t, t' in $\mathcal{S}$ with $t < t'$ such that $u \in C(t)$ and $v \in R(t)$ and $v \in L(t')$ and $u \in C(t')$. Then, u and v are adjacent in G by properties of permutation diagrams. If $l(u) = r(u)$ then u is not crossed by any minimal separation line in $\mathcal{S}$, and u and v are adjacent in G due to Lemma 3.5. Similarly, if $l(v) = r(v)$ then $uv \in E(G)$. Let $l(u) < r(u)$ and $l(v) < r(v)$. If $r(u) = l(v)$ then there are minimal separation lines s, s' in $\mathcal{S}$ with s' a successor of s such that $u \in C(s)$, $v \in R(s)$, $u \in L(s')$, $v \in C(s')$. Lemma 3.5 shows that u and v are adjacent in G . Finally, let $l(u) \leq l(v) < r(u) \leq r(v)$. Then, there is a minimal separation line t in $\mathcal{S}$ that crosses u and v . Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the enclosure of t . If $uv \notin E(G)$ then u and v are adjacent to a and d or to b and c in G , that are both pairs of non-adjacent vertices of $G(\mathcal{S})$. Thus, uv is the unique chord in a cycle of length 4 in $G(\mathcal{S})$, and $G(\mathcal{S})$ is a minimal triangulation of G .

Due to Lemma 3.2, $G(\mathcal{T})$ is a subgraph of $G(\mathcal{S})$, and since $G(\mathcal{T})$ is chordal due to Lemma 3.2, $G(\mathcal{S})$ and $G(\mathcal{T})$ are equal. Hence, $G(\mathcal{T})$ is a minimal triangulation of G . $\square$

4. A completeness result for minimal triangulations from permutation diagrams

In Section 3, we have defined an operation on permutation graphs that produces triangulations. The main result of that section, Theorem 3.6, presented a sufficient condition on the set of used scanlines so that the result of the operation is a minimal triangulation. In this section, we consider the complementary problem and ask whether every minimal triangulation can be the result of the operation. As the main result here we will obtain a positive answer. The proof of this result also relies on an important structural result about minimal triangulations of permutation graphs by Bodlaender, Kloks, Kratsch, that every minimal triangulation of a permutation graph is an interval graph [2]. This section is partitioned into two subsections. In the first subsection, we will study a special type of maximal clique in minimal triangulations and find a representation for them in permutation diagrams. In the second subsection, we will prove the main result.

4.1. Potential maximal cliques with inner vertices

The work of Bouchitté and Todinca showed the importance of the study of sets of vertices of a graph that form a maximal clique in a minimal triangulation [5,6]. Let G be a graph. A set M of vertices of G is called a *potential maximal clique* of G if there is a minimal triangulation H of G such that M is a maximal clique of H . Potential maximal cliques were introduced by Bouchitté and Todinca [5]. In this subsection, we are interested in a special type of potential maximal clique, namely the ones that are closed neighbourhoods of single vertices. A vertex x in a potential maximal clique M of G is called *inner vertex* (of M) if $N_G[x] = M$. In this case, we say that M has an inner vertex. We only mention here as a remark that potential maximal cliques with inner vertices do exist and every graph has such potential maximal cliques. The result of this subsection is a representation for potential maximal cliques with inner vertices of permutation graphs by a pair of minimal separation lines.

Lemma 4.1. Let G be a graph and let H be a minimal triangulation of G . Let x be a vertex that appears in exactly one maximal clique M of H . Then, x is an inner vertex of M .

Proof. Note that $M = N_H[x]$. By $G \subseteq H$, it follows that $N_G[x] \subseteq M$. Suppose that there is $v \in M$ with $v \neq x$ such that $vx \notin E(G)$. We consider $H' =_{\text{def}} H - xv$. Note that $N_{H'}[x]$ is a clique in H' . Hence, x cannot be vertex in a chordless cycle of length at least 4 in H' , and thus H' is chordal. We obtain a contradiction from $G \subseteq H' \subset H$. $\square$

Bouchitté and Todinca presented a characterisation of potential maximal cliques, that can be formulated for our case as in the following proposition.

Proposition 4.2 ([5]). *Let G be a graph and let x be a vertex of G . $N_G[x]$ is a potential maximal clique of G if and only if there is no vertex pair a, b with $a, b \in N_G(x)$ such that $N_G[x] \setminus \{a, b\}$ is an a, b -separator of G .*

We apply the characterisation of Proposition 4.2 to show a representation of potential maximal cliques of permutation graphs by using scanlines.

Lemma 4.3. *Let $G = G(\pi)$ be a permutation graph. Let M be a potential maximal clique of G that has an inner vertex. There is a pair s, t of minimal separation lines of G where t is a successor of s such that $M = C(s) \cup C(t) \cup (R(s) \cap L(t))$. The inner vertices of M are exactly the vertices in $R(s) \cap L(t)$.*

Proof. Let G have n vertices. By definition, there is a vertex x such that $N_G[x] = M$. Let

$$\begin{aligned} p_1 &=_{\text{def}} \max\{a < x : a \notin M\} \cup \{0\} \\ p_2 &=_{\text{def}} \min\{a > x : a \notin M\} \cup \{n+1\} \\ q_1 &=_{\text{def}} \max\{i < \pi^{-1}(x) : \pi(i) \notin M\} \cup \{0\} \\ q_2 &=_{\text{def}} \min\{i > \pi^{-1}(x) : \pi(i) \notin M\} \cup \{n+1\}. \end{aligned}$$

It holds that $p_1 < x < p_2$ and $\pi(q_1) <_\pi x <_\pi \pi(q_2)$. By construction, the vertices $p_1 + 1, \dots, p_2 - 1$ and $\pi(q_1 + 1), \dots, \pi(q_2 - 1)$ are contained in M . Thus, $s =_{\text{def}} (p_1 + \frac{1}{2}, q_1 + \frac{1}{2})$ and $t =_{\text{def}} (p_2 - \frac{1}{2}, q_2 - \frac{1}{2})$ are minimal separation lines of G , and $s < t$. By the definition of p_1, p_2, q_1, q_2 , it holds that $R(s) \cap L(t) \subseteq M$ and $C(s) \setminus C(t) \subseteq M$ and $C(t) \setminus C(s) \subseteq M$, and $C(s) \cap C(t) \subseteq M$ by properties of permutation diagrams. Since every vertex in M is contained in one of the four sets by the properties of permutation diagrams, we conclude that $M = C(s) \cup C(t) \cup (R(s) \cap L(t))$.

Next, we show that t is a successor of s . Assume the contrary and let s' be a minimal separation line of G with $s < s' < t$. Let $\left\langle \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right\rangle$ be the enclosure of s' . If s' does not cross x then, if $x \in R(s')$, a and c are not neighbours of x or, if $x \in L(s')$, b and d are not neighbours of x . Then, $a \leq p_1$ and $c \preceq_\pi \pi(q_1)$, or $p_2 \leq b$ and $\pi(q_2) \preceq_\pi d$. Both cases yield a contradiction to $s < s' < t$. Thus, $x \in C(s')$ and x has a pair of non-adjacent neighbours, a, d or b, c . It holds that $C(s')$ is an a, d - and a b, c -separator. It is not difficult to see that $C(s') \subseteq M$. Hence, $M \setminus \{a, d\}$ is an a, d -separator and $M \setminus \{b, c\}$ is a b, c -separator. This, however, contradicts the result of Proposition 4.2. Thus, t is a successor of s .

To conclude the lemma, it suffices to observe that the vertices in $R(s) \cap L(t)$ have the same closed neighbourhood due to Lemma 3.5. Since $x \in R(s) \cap L(t)$, every vertex in $R(s) \cap L(t)$ is an inner vertex of M . Every vertex in $C(s) \cup C(t)$ has a neighbour that is not contained in M , thus no vertex in $C(s) \cup C(t)$ can be inner vertex of M . $\square$

4.2. The structure of minimal triangulations

We show that every minimal triangulation of a permutation graph can be obtained by using a set of pairwise non-intersecting minimal separation lines. Hereby, we constitute the counterpart result of Theorem 3.6. Our proof is constructive in the sense that it provides an algorithm for the problem, given a minimal triangulation, finding an appropriate set of minimal separation lines. The construction works in several steps. The result is based on the strong structural result about minimal triangulations of permutation graphs by Bodlaender et al. [2].

Theorem 4.4 ([2]). *Minimal triangulations of permutation graphs are interval graphs.*

To give a brief outline of proof and construction, we are given a minimal triangulation H of a permutation graph $G = G(\pi)$. By Theorem 4.4, H is an interval graph, thus has a consecutive clique arrangement. The algorithm works as follows: (1) compute a consecutive clique arrangement for H that follows the vertex ordering of π , (2) find pairs of minimal separation lines for cliques with inner vertices, (3) add more minimal separation lines for other cliques. The second and third step are covered by the proof of Theorem 4.7. The first step is covered by the next lemma.

Lemma 4.5. *Let $G = G(\pi)$ be a permutation graph and let H be a minimal triangulation of G . Then, H has a consecutive clique arrangement $\langle A_1, \dots, A_k \rangle$ such that for every pair j, j' with $j < j'$ and every $y, y' \in V(G)$, if y is an inner vertex of A_j and y' is an inner vertex of $A_{j'}$ then $y <_\pi y'$.*

Proof. Let $\langle B_1, \dots, B_k \rangle$ be a consecutive clique arrangement for H . Note that B_1 and B_k have inner vertices. By the definition of inner vertex, it holds that inner vertices of different maximal cliques are pairwise non-adjacent in H and therefore in G . If $k = 1$ then $\langle B_1 \rangle$ trivially has the desired property. Let $k = 2$. Let Y_1 and Y_2 be the sets of inner vertices of respectively B_1 and B_2 . The properties of permutation diagrams easily show that if there are $u \in Y_1$ and $v \in Y_2$ with $u <_\pi v$ then $x <_\pi y$ for all vertex pairs x, y with $x \in Y_1$ and $y \in Y_2$. Thus, $\langle B_1, B_2 \rangle$ or $\langle B_2, B_1 \rangle$ has the desired property.

Henceforth, let $k \geq 3$. Let $\mathcal{I}$ be the set of indices $1 \leq i \leq k$ with B_i having an inner vertex. Let $\mathcal{Y}$ be a set that contains exactly one inner vertex from every B_i with $i \in \mathcal{I}$. By the properties of inner vertices, $\mathcal{Y}$ is an independent set of H and G . By the properties of permutation graphs, $\mathcal{Y}$ defines an increasing subsequence in π . Let $\mathcal{Y} = \{y_1, \dots, y_r\}$ where $y_1 < \dots < y_r$. It follows that $y_1 <_\pi \dots <_\pi y_r$. Let φ be a function such that $y_i \in B_{\varphi(i)}$ for every $1 \leq i \leq r$. We denote by $<_\varphi$ the

ordering over $\mathcal{I}$ that is implied by φ , i.e., that orders the vertices in $\mathcal{V}$ from left to right as they appear in $\langle B_1, \dots, B_k \rangle$. We show that we can reorder $B_1, \dots, B_k$ and still obtain a consecutive clique arrangement for H . In particular, we have to show that our re-arrangement results in an ordering that has the consecutive ones property. Our construction works in steps and incrementally constructs a consecutive clique arrangement for H .

Claim A. $\langle B_{\varphi(1)}, \dots, B_{\varphi(r)} \rangle$ has the consecutive ones property.

Proof. Let x be a vertex of G . Let i, i' be such that $i < i'$ and $x \in B_{\varphi(i)} \cap B_{\varphi(i')}$. It holds that x is adjacent to y_i and $y_{i'}$ in G due to Lemma 4.1. It holds that $y_i < y_{i+1} \leq y_{i'}$ and $y_i <_{\pi} y_{i+1} <_{\pi} y_{i'}$. Thus, x is adjacent to y_{i+1} in G and therefore in H . Hence, $x \in B_{\varphi(i+1)}$. Iterative application of this argument shows that $x \in B_{\varphi(i)} \cap \dots \cap B_{\varphi(i')}$. $\square$

We use the arrangement in Claim A as starting point and incrementally fill the maximal cliques of H that have no inner vertex. Let $1 \leq j, j' \leq r$ with $j <_{\varphi} j'$. We call the pair $[j, j']$ *consecutive* if there is no $i \in \mathcal{I}$ such that $j <_{\varphi} i <_{\varphi} j'$. A consecutive pair $[j, j']$ is called *connected* if $|\varphi(j) - \varphi(j')| \geq 2$. The goal is to place $B_{\varphi(j)+1}, \dots, B_{\varphi(j')-1}$ between $B_{\varphi(j)}$ and $B_{\varphi(j')}$. These maximal cliques exist only in case of connected consecutive pairs. A difficulty for placing these maximal cliques arises in case $|j - j'| > 1$. Let $M_{j,j'}^+ =_{\text{def}} B_{\varphi(j)} \cap B_{\varphi(j)+1}$ and $M_{j,j'}^- =_{\text{def}} B_{\varphi(j')-1} \cap B_{\varphi(j')}$. We show two properties.

Property 1. Let $1 \leq j, j' \leq r$ with $j <_{\varphi} j'$ and $[j, j']$ consecutive. Let there be $i \in \mathcal{I}$ with $j < i < j'$ or $j' < i < j$. If $j' <_{\varphi} i$ then $M_{j,j'}^- \subseteq B_{\varphi(i)}$, if $i <_{\varphi} j$ then $M_{j,j'}^+ \subseteq B_{\varphi(i)}$.

Proof. Since the two cases are symmetric, we consider the case that $j' <_{\varphi} i$. Let x be a vertex in $M_{j,j'}^-$ and suppose that $x \notin B_{\varphi(i)}$. This means that there is a $y_j, y_{j'}$ -path P in H that contains no neighbour of y_i . Since H is a minimal triangulation of G we can choose P as such that it is a path also in G : if x is a neighbour also of y_j then $(y_j, x, y_{j'})$ is such a path, otherwise there is a smallest index r such that B_r contains a neighbour u of x and $u \notin B_{r+1}$; this is true since B_r is a maximal clique of H without inner vertex. Since x and u occur together only in B_r , xu is an edge of G due to H being a minimal triangulation of G . Continuing the construction with u yields the desired path of G . However, this is not possible according to the assumptions $y_j < y_i < y_{j'}$ and $y_j <_{\pi} y_i <_{\pi} y_{j'}$. $\square$

Property 2. Let $1 \leq j, j' \leq r$ with $j <_{\varphi} j'$ and $[j, j']$ connected consecutive. Let there be $i, i' \in \mathcal{I}$ with $i <_{\varphi} j <_{\varphi} j' <_{\varphi} i'$. If $j < i, i' < j'$ then $i < i'$, if $j' < i, i' < j$ then $i' < i$.

Proof. By a symmetry argument, it suffices to consider the case that $j < i, i' < j'$. By the assumption $\varphi(j') - \varphi(j) \geq 2$, it holds that there is a vertex in $M_{j,j'}^+$ that is not in $M_{j,j'}^-$ and vice versa. Therefore, it follows from $i <_{\varphi} j <_{\varphi} j' <_{\varphi} i'$ and the result of Property 1 that y_j and $y_{i'}$ have a common neighbour that is not neighbour of $y_{i'}$, and $y_{j'}$ and $y_{i'}$ have a common neighbour that is not neighbour of y_i . Then, $y_j < y_{i'} < y_i < y_{j'}$ and $y_j <_{\pi} y_{i'} <_{\pi} y_i <_{\pi} y_{j'}$ would result in a contradiction.

Following Property 2, there is a unique number $\psi(j, j')$ between j and j' that separates the maximal cliques with inner vertices between B_j and $B_{j'}$ in the arrangement of Claim A. For a formal definition, we have to distinguish between two cases. First, let $j < j'$. Then, $\psi(j, j')$ is the number with $j \leq \psi(j, j') < j'$ such that for all $i \in \mathcal{I}$, if $j < i \leq \psi(j, j')$ then $i <_{\varphi} j$, if $\psi(j, j') < i < j'$ then $j' <_{\varphi} i$. Second, let $j' < j$. Then, $\psi(j, j')$ is the number with $j' \leq \psi(j, j') < j$ such that for all $i \in \mathcal{I}$, if $j' < i \leq \psi(j, j')$ then $j' <_{\varphi} i$, if $\psi(j, j') < i < j$ then $i <_{\varphi} j$.

Suppose that there are $1 \leq p, p' \leq r$ with $p <_{\varphi} p'$ and $[p, p']$ connected consecutive such that $\psi(p, p') = \psi(j, j')$. We show that this yields a contradiction. We have to consider four different general cases where two cases reduce to the two other ones by simply reversing the given consecutive clique arrangement for H . Thus, we can restrict to the following two cases, each of which implies two subcases.

assumption	w.l.o.g.	two cases
$j < j'$ and $p < p'$	$j \leq p$	$j \leq p \leq \psi(j, j') < p' \leq j'$ $j < p \leq \psi(j, j') < j' < p'$
$j < j'$ and $p' < p$	$j \leq p'$	$j \leq p' \leq \psi(j, j') < p \leq j'$ $j < p' \leq \psi(j, j') < j' < p$

We consider the first case. In the first subcase, it holds that $p <_{\varphi} j <_{\varphi} j' <_{\varphi} p'$. This is a contradiction to $[p, p']$ being consecutive. In the second subcase, it holds that $p <_{\varphi} j$ and $p' <_{\varphi} j'$. By the consecutiveness assumption, it follows that $p <_{\varphi} p' <_{\varphi} j <_{\varphi} j'$. It holds that $M_{j,j'}^+ \subseteq B_{\varphi(p)}$ and $M_{p,p'}^- \subseteq B_{\varphi(j')}$ due to Property 1. This particularly means that $M_{j,j'}^+ = M_{p,p'}^-$. Since $B_{\varphi(j)+1}$ has no inner vertex, there is a vertex x in $B_{\varphi(j)+1}$ that is not contained in $B_{\varphi(j)+2}$ and therefore x is not contained in $B_{\varphi(j')}$. However, since x is a vertex in $M_{j,j'}^+ = M_{p,p'}^-$, this is not possible. Similarly for $M_{p,p'}^-$. Hence, the first case cannot happen. For the second case, the first subcase is not possible, since it implies $p' <_{\varphi} p$. Thus, similar to the second subcase of the first case, it holds that $p <_{\varphi} p' <_{\varphi} j <_{\varphi} j'$ and we obtain a contradiction with the same arguments as above.

We fill the remaining maximal cliques of H into the arrangement of Claim A. We show that

$$\langle B_{\varphi(1)}, \dots, B_{\varphi(j)}, \dots, B_{\varphi(\psi(j, j'))}, B_{\varphi(j)+1}, \dots, B_{\varphi(j')-1}, B_{\varphi(\psi(j, j')+1)}, \dots, B_{\varphi(j')}, \dots, B_{\varphi(r)} \rangle$$

or

$$\langle B_{\varphi(1)}, \dots, B_{\varphi(j')}, \dots, B_{\varphi(\psi(j,j'))}, B_{\varphi(j')-1}, \dots, B_{\varphi(j)+1}, B_{\varphi(\psi(j,j')+1)}, \dots, B_{\varphi(j)}, \dots, B_{\varphi(r)} \rangle$$

has the consecutive ones property, depending on whether $j < j'$ or $j' < j$. The proof is not difficult and mainly relies on Properties 1 and 2 and the definition of $\psi(j, j')$. We only consider the first case, $j < j'$; the second case can be proved analogously. Since we started from arrangements with the consecutive ones property, it remains to check for the following three sets:

- $B_{\varphi(j)} \cap B_{\varphi(j)+1}$
this set is equal to $M_{j,j'}^+$ and therefore contained in $B_{\varphi(j)}, \dots, B_{\varphi(\psi(j,j'))}$ due to Property 1
- $B_{\varphi(j')-1} \cap B_{\varphi(j')}$
this set is equal to $M_{j,j'}^-$ and therefore contained in $B_{\varphi(\psi(j,j')+1)}, \dots, B_{\varphi(j')}$ due to Property 1
- $B_{\varphi(\psi(j,j'))} \cap B_{\varphi(\psi(j,j')+1)}$
this set is contained in $B_{\varphi(j)} \cap B_{\varphi(j')}$ and therefore contained in $B_{\varphi(j)+1}, \dots, B_{\varphi(j')-1}$.

Hence, one of the arrangements has the consecutive ones property. And since the vertices in $(B_{\varphi(j)+1} \cup \dots \cup B_{\varphi(j')-1}) \setminus (B_{\varphi(j)} \cup B_{\varphi(j')})$ appear in no other maximal cliques of H , we can finally conclude that filling the remaining maximal cliques in the described way yields a consecutive clique arrangement for H with the desired properties. $\square$

Lemma 4.6. Let $G = G(\pi)$ be a permutation graph. Let $\mathcal{S} = \{s_1, \dots, s_k\}$ be a set of pairwise non-intersecting minimal separation lines of G with $s_1 = \mathbf{0}$ and $s_k = \mathbf{1}$, where $\mathbf{0} < s_2 < \dots < s_{k-1} < \mathbf{1}$. Then, $G(\mathcal{S})$ has a consecutive clique arrangement $\langle A_1, \dots, A_{k-1} \rangle$ where $A_i = (R(s_i) \cap L(s_{i+1})) \cup C(s_i) \cup C(s_{i+1})$ for every $1 \leq i < k$ and $A_{i-1} \cap A_i = C(s_i)$ for every $1 < i < k$.

Proof. Let (l, r) be the endpoint function pair of $\mathcal{S}$. Note that $2 \leq l(x) \leq r(x) \leq k$ for every vertex x of G . For every $1 \leq i \leq k-1$, let A_i be the set of vertices x with $l(x) \leq i+1 \leq r(x)$. We show that $\langle A_1, \dots, A_{k-1} \rangle$ satisfies the lemma. By the definition of $G(\mathcal{S})$ it follows immediately that $A_1, \dots, A_{k-1}$ are cliques of $G(\mathcal{S})$ and that $\langle A_1, \dots, A_{k-1} \rangle$ has the consecutive ones property. Since the intervals in $\mathcal{F}_G(\mathcal{S})$ have only integer endpoints, every maximal clique of $G(\mathcal{S})$ appears as A_i for some i .

For every vertex x of G and every $1 \leq i \leq k-1$, the following holds

$$\begin{aligned} x \in R(s_i) \cap (L(s_{i+1}) \cup C(s_{i+1})) &\iff l(x) = i+1 \\ x \in L(s_{i+1}) \cap (R(s_i) \cup C(s_i)) &\iff i+1 = r(x) \\ x \in R(s_i) \cap L(s_{i+1}) &\iff l(x) = i+1 = r(x) \\ x \in C(s_i) &\iff l(x) < i+1 \leq r(x). \end{aligned}$$

Since $l(x) < i+1 \leq r(x)$ is equivalent to $x \in A_{i-1} \cap A_i$, we conclude the representation properties for A_i and $A_{i-1} \cap A_i$.

It remains to show that each A_i is a maximal clique of $G(\mathcal{S})$. Let $1 \leq i < k$. To show that A_i is a maximal clique of $G(\mathcal{S})$, it suffices to show that there are (not necessarily different) vertices x and y with $r(x) = l(y) = i+1$. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ be the enclosure of respectively s_i and s_{i+1} . It holds that $b, d \in R(s_i)$ and $a', c' \in L(s_{i+1})$. Since s_i and s_{i+1} can have at most one endpoint in common, $b \in R(s_i) \setminus R(s_{i+1})$ or $d \in R(s_i) \setminus R(s_{i+1})$. Thus, $l(b) = i+1$ or $l(d) = i+1$. Similarly, $r(a') = i+1$ or $r(c') = i+1$. Hence, A_i is a maximal clique of $G(\mathcal{S})$. $\square$

An important consequence of Lemma 4.6 is that the set of vertices that is defined by two minimal separation lines is invariant with respect to minimal separation lines to the left or right. This will be important for the proof of the next theorem and its iterative approach.

Theorem 4.7. Let $G = G(\pi)$ be a permutation graph and let H be a minimal triangulation of G . There is a set $\mathcal{S}$ of pairwise non-intersecting minimal separation lines of G such that $G(\mathcal{S}) = H$.

Proof. Due to Theorem 4.4, H is an interval graph. If H is a complete graph then G is a complete graph, with $\mathbf{0}$ and $\mathbf{1}$ the only minimal separation lines of G . Since $\mathbf{0} < \mathbf{1}$, $\{\mathbf{0}, \mathbf{1}\}$ is a set of pairwise non-intersecting minimal separation lines of G such that $G(\{\mathbf{0}, \mathbf{1}\}) = H$. Note that the equivalence is trivial with H and G being complete and $G \subseteq G(\{\mathbf{0}, \mathbf{1}\})$ due to Lemma 3.2. Henceforth, assume that H and G are not complete. Let $\langle A_1, \dots, A_k \rangle$ be a consecutive clique arrangement for H that has the property of Lemma 4.5, where $k \geq 2$. According to definition, $A_1, \dots, A_k$ are potential maximal cliques of G . Let $\mathcal{I}$ be the set of indices $1 \leq i \leq k$ with A_i having an inner vertex. Let $\mathcal{I} = \{i_1, \dots, i_r\}$ where $1 = i_1 < \dots < i_r = k$. For every $i \in \mathcal{I}$, let s_i and s'_i be minimal separation lines of G where s'_i is a successor of s_i such that $A_i = C(s_i) \cup C(s'_i) \cup (R(s_i) \cap L(s'_i))$; s_i and s'_i exist due to Lemma 4.3. Let $\mathcal{S}_0 =_{\text{def}} \{s_i, s'_i : i \in \mathcal{I}\}$. We show three properties of $\mathcal{S}_0$.

Property 1. $s_{i_1} < s'_{i_1} \leq s_{i_2} < \dots < s'_{i_r}$.

Proof. Suppose that there are scanlines $s, t \in \mathcal{S}_0$ that intersect. Let $j, j' \in \mathcal{I}$ with $s \in \{s_j, s'_j\}$ and $t \in \{s_{j'}, s'_{j'}\}$. Since $s_i < s'_i$ for every $i \in \mathcal{I}$, $j \neq j'$. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the enclosure of t . Depending on how s and t intersect, $a, d \in C(s)$ or $b, c \in C(s)$.

Let $y \in R(s_j) \cap L(s'_j)$, which is an inner vertex of A_j due to Lemma 4.3. According to statement 2 of Lemma 3.5, y is adjacent to a and d or to b and c . Hence, $y \in C(t)$, which would imply $y \in A_{j'}$ and therefore contradict y being an inner vertex of A_j . We conclude that $\mathcal{S}_0$ is a set of pairwise non-intersecting scanlines of G . The claim follows with the properties of $\langle A_1, \dots, A_k \rangle$. $\square$

Suppose that there is a vertex x with $x \in L(s_1)$. Then, x is not inner vertex in a maximal clique of H or adjacent to an inner vertex. So, $x \in (A_{j+1} \cup \dots \cup A_{i_{j+1}-1}) \setminus (A_{ij} \cup A_{i_{j+1}})$ for some $1 \leq j < r$. With y and y' inner vertices in respectively A_{ij} and $A_{i_{j+1}}$ and since H is a minimal triangulation of G , there is an x, y -path in H and G without a neighbour of y' , and there is an x, y' -path in H and G without a neighbour of y . However, this is not possible since both paths contain neighbours of the inner vertices of A_{j_1} by the properties of permutation diagrams. Hence, $L(s_1) = \emptyset$. Similarly, $R(s'_k) = \emptyset$. And since s_1 and s'_k are minimal separation lines of G , $s_1 = \mathbf{0}$ and $s'_k = \mathbf{1}$.

Property 2. For every $1 \leq j < r$, $R(s'_j) \cap L(s_{j+1}) = (A_{j+1} \cup \dots \cup A_{i_{j+1}-1}) \setminus (A_{ij} \cup A_{i_{j+1}})$.

Proof. Let $1 \leq j < r$. Let y and y' be inner vertices in respectively A_{ij} and $A_{i_{j+1}}$. We show two inclusions. Let $x \in R(s'_j) \cap L(s_{j+1})$. Then, $y < x < y'$ and $y <_\pi x <_\pi y'$, which particularly means that x cannot be inner vertex for any A_i . Suppose that $x \notin (A_{j+1} \cup \dots \cup A_{i_{j+1}-1}) \setminus (A_{ij} \cup A_{i_{j+1}})$, which means that $x \in (A_1 \cup \dots \cup A_{i_{j-1}}) \setminus A_{ij}$ or $x \in (A_{i_{j+1}+1} \cup \dots \cup A_k) \setminus A_{i_{j+1}}$. Since the two cases are symmetric, we restrict to the former case, and therefore $i_j \geq j \geq 2$. Let i' be smallest such that $x \in A_{i'}$, and let j' be largest such that $i_j \leq i'$. Note that $j' < j$. Let z be inner vertex in $A_{i_{j'}}$. Since H is a minimal triangulation of G , there is a z, x -path in H and G that contains no vertex from $A_{ij} \cup \dots \cup A_k$, in particular, no neighbour of y . This yields a contradiction to $z < y < x$ and $z <_\pi y <_\pi x$ and the properties of permutation diagrams. The converse inclusion follows from the fact that the two sides of the equivalence define partitions of $V(G) \setminus (A_{i_1} \cup \dots \cup A_{i_r})$ with the same number of classes and the above inclusion result. $\square$

Property 3. For every $1 \leq j < r$ with $i_{j+1} - i_j \geq 2$, $C(s'_j) = A_{ij} \cap A_{i_{j+1}}$ and $C(s_{j+1}) = A_{i_{j+1}} \cap A_{i_{j+1}-1}$.

Proof. Let $1 \leq j < r$ with $i_{j+1} - i_j \geq 2$. Because of symmetry, it suffices to show one of the two cases. We show that $C(s'_j) = A_{ij} \cap A_{i_{j+1}}$ by showing the two inclusions. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the enclosure of s'_j . We show that $b, d \in A_{i_{j+1}} \cup \dots \cup A_{i_{j+1}}$. It holds that $b, d \in R(s'_j)$, and therefore, $b, d \notin A_{ij}$. Let y' be an inner vertex of $A_{i_{j+1}}$. If b is not adjacent to y' in G then $b < y'$ and $b <_\pi y'$, which means that $b \in L(s_{j+1})$. By Property 2, it follows that $b \in A_{j+1} \cup \dots \cup A_{i_{j+1}}$. Similarly for d . Let $x \in C(s'_j)$. By the choice of s'_j , $x \in A_{ij}$. By the properties of permutation diagrams, x is adjacent to b or d , and thus, $x \in A_{i_{j+1}}$ due to the consecutive ones property. For the converse, let $x \in A_{ij} \cap A_{i_{j+1}}$. Let y be an inner vertex of A_{ij} . Let i' be largest possible such that $x \in A_{i'}$. Clearly, $i' \geq i_j + 1$. Let z be a vertex in $A_{i'} \setminus A_{i'-1}$. By Property 2 and previous results, $z \in R(s'_j)$. It holds that $x \in N_H[y] \cap N_H[z]$, and since H is a minimal triangulation of G , $x \in N_G[y] \cap N_G[z]$. Hence, x has a neighbour in $L(s'_j)$ and in $R(s'_j)$, which means that $x \in C(s'_j)$. $\square$

We extend set $\mathcal{S}_0$ by adding more minimal separation lines. Let $\mathcal{R}_0 =_{\text{def}} \{1, \dots, k\} \setminus \mathcal{I}$ and let $t =_{\text{def}} |\mathcal{R}_0|$. We inductively define sets of minimal separation lines of G , $\mathcal{S}_1, \dots, \mathcal{S}_t$, with $\mathcal{S}_0 \subseteq \dots \subseteq \mathcal{S}_t$, and sets $\mathcal{R}_1, \dots, \mathcal{R}_t$, with $\mathcal{R}_0 \supset \dots \supset \mathcal{R}_t$, such that for every $0 \leq i \leq t$ holds: for every $j \in \{1, \dots, k\} \setminus \mathcal{R}_i$, A_j is a maximal clique of $G(\mathcal{S}_i)$. We show that $\mathcal{S}_0, \dots, \mathcal{S}_t$ have Properties 1, 2, 3. Due to Property 1 and Lemmas 4.3 and 4.6, the claims hold for $\mathcal{S}_0$ and $\mathcal{R}_0$. Let $1 \leq i \leq t$ and assume that $\mathcal{S}_{i-1}$ and $\mathcal{R}_{i-1}$ have already been defined and shown to satisfy the claim. Let $j =_{\text{def}} \min \mathcal{R}_{i-1}$, and let $\mathcal{R}_i =_{\text{def}} \mathcal{R}_{i-1} \setminus \{j\}$. We distinguish two basic cases. As the first case, let $j + 1 \in \mathcal{I}$. Let $\mathcal{S}_i =_{\text{def}} \mathcal{S}_{i-1}$. We consider the scanlines s'_{j-1} and s_{j+1} . Note that s_{j+1} is the left scanline chosen for A_{j+1} in $\mathcal{S}_0$, and s'_{j-1} is either the right scanline of A_{j-1} in $\mathcal{S}_0$ or it is the scanline added in step $i - 1$ to obtain $\mathcal{S}_{i-1}$. By induction hypothesis, $C(s'_{j-1}) = A_{j-1} \cap A_j$ and $C(s_{j+1}) = A_{j+1} \cap A_j$. Since A_j has no inner vertex and therefore $A_j = (A_{j-1} \cap A_j) \cup (A_{j+1} \cap A_j)$ and since $R(s'_{j-1}) \cap L(s_{j+1}) = \emptyset$ by induction hypothesis, we obtain $A_j = (R(s'_{j-1}) \cap L(s_{j+1})) \cup C(s'_{j-1}) \cup C(s_{j+1})$.

As the second case, let $j + 1 \notin \mathcal{I}$. In this case, we choose a minimal separation line that we add to $\mathcal{S}_{i-1}$ and show the necessary properties. The following arguments are best understood in the permutation diagram of G . Consider $B =_{\text{def}} A_j \setminus A_{j+1}$. Let p, q be such that $s'_{j-1} = (p, q)$. By induction hypothesis, $B \subseteq C(s'_{j-1})$. Suppose that there are vertices x, x' in B with $x < p$ and $x' > p$. Let z be a vertex in $A_j \setminus A_{j-1}$. It holds that $x, x' \in C(s'_{j-1})$ and $z \notin C(s'_{j-1})$ and $xz, x'z \in E(G)$. Let $j' \in \mathcal{I}$ be smallest with $j < j'$, and let y be an inner vertex of $A_{j'}$. Since z is not inner vertex and is contained in A_{j+1} , there is a y, z -path in H and G that contains no vertex from $A_1 \cup \dots \cup A_{j-1}$. However, since y is non-adjacent to x and x' in H and therefore in G , the situation in the permutation diagram shows that every y, z -path in G contains a neighbour of x or x' besides z , which yields a contradiction. Therefore, for all vertex pairs x, x' from B , $p < x$ if and only if $p < x'$. Let $u, v \in B$ be such that $x \leq u$ and $x \leq_\pi v$ for all $x \in B$. Let $p' =_{\text{def}} u + \frac{1}{2}$ and $q' =_{\text{def}} \pi^{-1}(v) + \frac{1}{2}$. We show that (p', q) or (p, q') satisfies the claims, depending on whether $p < u$ or $q < \pi^{-1}(v)$. In the following, we assume the former case; the latter case is symmetric and can be proved analogously. Let $s'_j =_{\text{def}} (p', q)$. We show that s'_j satisfies the conditions. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the enclosure of s'_j . Since c is an enclosure vertex of s'_{j-1} and since $a = u$, it holds that $a, c \in L(s'_j)$. Suppose that $b \in C(s'_j)$.

Then, for all $x \in R(s'_{j-1})$, $xa \in E(G)$ if and only if $xb \in E(G)$. However, since $b \notin B$ and therefore has a neighbour in G that is in $A_{j+1} \cup \dots \cup A_k$ and by induction hypothesis in $R(s'_{j-1})$ and that is non-adjacent to a , we obtain a contradiction. Thus, $b \notin C(s'_j)$. Finally, suppose that $d \in C(s'_j)$. Then, a and d are adjacent in G , so that $d \in A_j$. And since $d \notin A_1 \cup \dots \cup A_{j-1}$ by induction hypothesis and since A_j has no inner vertex, d has a neighbour in $A_{j+1} \cup \dots \cup A_k$ that is non-adjacent to a in H . By assumption, this neighbour is a vertex in $R(s'_{j-1})$. This, however, is not possible in a permutation diagram, so that we conclude that s'_j is a minimal separation line of G . Next, suppose that s'_j intersects with $s_{j'}$. Then, s'_j also intersects with y , which particularly means that a and y are adjacent in G and therefore in H . This, however, is not the case. Let $\delta_j =_{\text{def}} \delta_{j-1} \cup \{s'_j\}$. We conclude that δ_j is a set of pairwise non-intersecting minimal separation lines of G . It remains to verify Properties 2 and 3. Note that by the definition of s'_j , $R(s'_{j-1}) \cap L(s'_j)$ is empty. We show that $C(s'_j) = A_j \cap A_{j+1}$. Let $x \in C(s'_j)$. If $x \in C(s'_{j-1})$ then $x \in (A_{j-1} \cap A_j) \setminus B$ and therefore $x \in A_{j+1}$. If $x \notin C(s'_{j-1})$ then $x \in R(s'_{j-1})$ and $xa \in E(G)$. Thus, $x \in A_j$ and since A_j has no inner vertex, $x \in A_{j+1}$. Hence, $C(s'_j) \subseteq A_j \cap A_{j+1}$. For the converse, let $x \in A_j \cap A_{j+1}$. If $x \notin A_{j-1}$ then x and a are adjacent in G , and since $x \notin C(s'_{j-1}) \cup L(s'_{j-1})$, $x \in R(s'_{j-1})$ and thus $x \in C(s'_j)$. If $x \in A_{j-1}$ then $x \in C(s'_{j-1})$ and since x has a neighbour in $R(s'_j)$, $x \in C(s'_j)$. We conclude that $C(s'_j) = A_j \cap A_{j+1}$. In particular, we conclude that $A_j = (R(s'_{j-1}) \cap L(s'_j)) \cup C(s'_{j-1}) \cup C(s'_j)$. And since $R(s'_{j-1}) \cap L(s'_j) = (A_j \cup \dots \cup A_{j'-1}) \setminus (A_{j-1} \cup A_{j'})$, we conclude that $R(s'_j) \cap L(s_{j'}) = (A_{j+1} \cup \dots \cup A_{j'-1}) \setminus (A_j \cup A_{j'})$.

By the above construction and the result of [Lemma 4.6](#), we obtain δ_t with the property that every maximal clique of H is a maximal clique of $G(\delta_t)$. Hence, $H \subseteq G(\delta_t)$. Suppose that $G(\delta_t)$ has a maximal clique A that is not a maximal clique of H . By construction of δ_t , there is j such that $j+1 \in \mathcal{I}$ and $A = (R(s'_j) \cap L(s_{j+1})) \cup C(s'_j) \cup C(s_{j+1})$. Since no vertex from $R(s'_j) \cap L(s_{j+1})$ can appear in another maximal clique of $G(\delta_t)$, it must hold that this set is empty. With the properties of maximal cliques without inner vertex, there is a pair of vertices in A that are adjacent in G and that do not appear together in any other maximal clique of $G(\delta_t)$. This particularly means that H cannot contain this edge of G , which is a contradiction to H being a triangulation of G . Therefore, A does not exist, $H = G(\delta_t)$, and we finally conclude the theorem. $\square$

Corollary 4.8. *Let $G = G(\pi)$ be a permutation graph. An interval graph H is a minimal triangulation of G if and only if there is a maximal set $\mathcal{S}$ of pairwise non-intersecting minimal separation lines of G with $G(\mathcal{S}) = H$.*

Proof. Let H be an interval graph. If there is a maximal set $\mathcal{S}$ of pairwise non-intersecting minimal separation lines of G with $G(\mathcal{S}) = H$ then H is a minimal triangulation of G due to [Theorem 3.6](#). For the converse, let H be a minimal triangulation of G . Due to [Theorem 4.7](#), there is a set $\mathcal{T}$ of pairwise non-intersecting minimal separation lines of G with $G(\mathcal{T}) = H$. Let $\mathcal{S}$ be a maximal set of pairwise non-intersecting minimal separation lines of G with $\mathcal{T} \subseteq \mathcal{S}$. Note that $\mathcal{S}$ exists and can be constructed efficiently by simply adding further minimal separation lines. Due to [Lemma 3.2](#), $G \subseteq G(\mathcal{S}) \subseteq G(\mathcal{T})$, and since $G(\mathcal{S})$ is chordal, it follows $G(\mathcal{S}) = G(\mathcal{T}) = H$ by H being a minimal triangulation of G . $\square$

[Corollary 4.8](#) establishes a correspondence between minimal triangulations and maximal sets of pairwise non-intersecting minimal separation lines. As the following lemma shows that two different such scanline sets define different triangulations, the correspondence of [Corollary 4.8](#) is 1-to-1.

Lemma 4.9. *Let $G = G(\pi)$ be a permutation graph. Let $\mathcal{S}$ and $\mathcal{T}$ be two maximal sets of pairwise non-intersecting minimal separation lines of G . It holds that $G(\mathcal{S}) = G(\mathcal{T})$ if and only if $\mathcal{S} = \mathcal{T}$.*

Proof. Clearly, if $\mathcal{S} = \mathcal{T}$ then $G(\mathcal{S}) = G(\mathcal{T})$. For the converse, let $\mathcal{S} \neq \mathcal{T}$. By the maximality assumption, there are $s \in \mathcal{S}$ and $t \in \mathcal{T}$ such that s and t intersect. Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the enclosure of t . Depending on the way how s and t intersect, $b, c \in C(s)$ or $a, d \in C(s)$. It holds that bc and ad are not edges of $G(\mathcal{T})$ but one of the two edges is an edge of $G(\mathcal{S})$. Hence, $G(\mathcal{S}) \neq G(\mathcal{T})$. $\square$

5. Minimal triangulation problems in linear time

In the first part of the paper, we have developed a representation of minimal triangulations of permutation graphs. The results culminated in the characterisation of [Corollary 4.8](#). In the second part of the paper, we will apply this characterisation to devise linear-time algorithms for various problems about minimal triangulations. In particular, we will give linear-time algorithms for computing treewidth and minimum fill-in for permutation graphs. The outline of the algorithms is as follows: (1) we define an acyclic digraph with weights on vertices and arcs, (2) we solve a shortest-path problem on the digraph. The digraph represents all maximal sets of pairwise non-intersecting minimal separation lines of a permutation graph and can be computed in linear time. Every considered minimal triangulation problem can be formulated as a shortest-path problem on the digraph and can be solved in linear time. This section is partitioned into two subsections, the first one presenting the theoretical results, the second one presenting the algorithms.

5.1. Problems and solutions

Let $G = G(\pi)$ be a permutation graph. The *minimal separation lines graph* of G , denoted as $Sl(\pi)$, is a digraph with vertex and arc set as follows:

- + $Sl(\pi)$ has a vertex for every minimal separation line of G ; each vertex is labelled with the corresponding minimal separation line

+ (u, v) , with s_u and s_v being the minimal separation lines of G that respectively u and v are labelled with, is an arc of $\text{Sl}(\pi)$ if and only if s_v is a successor of s_u .

The minimal separation lines graph represents all maximal sets of pairwise non-intersecting minimal separation lines of G .

Lemma 5.1. *Let $G = G(\pi)$ be a permutation graph. The following holds:*

- (1) $\text{Sl}(\pi)$ is acyclic
- (2) $\text{Sl}(\pi)$ has a unique source and a unique sink vertex
- (3) let $\mathcal{S} = \{s_1, \dots, s_k\}$ be a set of minimal separation lines of G and let $x_1, \dots, x_k$ be the corresponding vertices of $\text{Sl}(\pi)$, where x_i is labelled with s_i , $1 \leq i \leq k$. It holds that $\mathcal{S}$ is a maximal set of pairwise non-intersecting minimal separation lines where $s_1 < \dots < s_k$ if and only if $(x_1, \dots, x_k)$ is a source–sink path of $\text{Sl}(\pi)$.

Proof. The three statements directly follow from the definition of successor for minimal separation lines and the properties of scanlines **0** and **1**. $\square$

For a digraph G , a *topological ordering* for G is a vertex ordering $\tau = \langle u_1, \dots, u_n \rangle$ for G such that for every arc (u, v) of G , $u \prec_\tau v$. It is well known that acyclic digraphs have topological orderings, and they can be computed in linear time. As a corollary of the first statement of Lemma 5.1, $\text{Sl}(\pi)$ has a topological ordering.

Lemma 5.2. *Let $G = G(\pi)$ be a permutation graph. Let $\text{Sl}(\pi)$ be given. Then, the following three problems can be solved in time linear in $\text{Sl}(\pi)$:*

- compute the smallest number of maximal cliques in a minimal triangulation of G
- compute the largest number of maximal cliques in a minimal triangulation of G
- compute the number of minimal triangulations of G .

Proof. By the 1-to-1 correspondence between minimal triangulations of G and source–sink paths of $\text{Sl}(\pi)$ due to Corollary 4.8 and Lemma 4.9, the three problems can be reformulated as: find a source–sink path of $\text{Sl}(\pi)$ of smallest length, find such a path of largest length, compute the number of source–sink paths. For the first two problems, it suffices to observe that for a source–sink path P of $\text{Sl}(\pi)$, with $\mathcal{S}$ the set of scanlines the vertices of P are labelled with in $\text{Sl}(\pi)$, the length of P is equal to the number of maximal cliques of $G(\mathcal{S})$ due to Lemma 4.6. By processing the vertices of $\text{Sl}(\pi)$ according to a topological ordering, the three problems can be solved in time linear in $\text{Sl}(\pi)$. $\square$

We add weights to $\text{Sl}(\pi)$ and define two weighted digraphs. The *weighted minimal separation lines graph* of G , denoted as $\text{Sl}^+(\pi)$, is the digraph $\text{Sl}(\pi)$ with vertex and arc weights as follows:

- for a vertex u of $\text{Sl}^+(\pi)$ with minimal separation line label s , the weight of u is $|\mathcal{C}(s)|$
- for an arc (u, v) of $\text{Sl}^+(\pi)$ where u and v are labelled with minimal separation lines s and t , respectively, the weight of (u, v) is $|\mathcal{C}(s) \cap \mathcal{C}(t)|$.

The *maximum arc weight* of a path of $\text{Sl}^+(\pi)$ is the maximum taken over the weights of the arcs of the path. The *binomial-weighted minimal separation lines graph* of G , denoted as $\text{Sl}^*(\pi)$, is obtained from $\text{Sl}^+(\pi)$ by replacing every weight a by $\binom{a}{2}$. Let P be a path of $\text{Sl}^*(\pi)$. The *arc-minus-vertex weight* of P is the sum of the arc weights minus the sum of the vertex weights on P . With these definitions, treewidth and minimum fill-in become shortest-path problems.

Lemma 5.3. *Let $G = G(\pi)$ be a permutation graph.*

- (1) *The treewidth of G is equal to the minimum taken over the maximum arc weights of the source–sink paths of $\text{Sl}^+(\pi)$ minus 1. If $\text{Sl}^+(\pi)$ is given, $\text{tw}(G)$ and a source–sink path of $\text{Sl}^+(\pi)$ with minimum maximum arc weight among the source–sink paths can be computed in time linear in $\text{Sl}^+(\pi)$.*
- (2) *The minimum fill-in of G is equal to the minimum taken over the arc-minus-vertex weight of the source–sink paths of $\text{Sl}^*(\pi)$ minus the number of edges of G . If $\text{Sl}^*(\pi)$ is given, $\text{mfi}(G)$ and a source–sink path of $\text{Sl}^*(\pi)$ with minimum arc-minus-vertex weight among the source–sink paths of $\text{Sl}^*(\pi)$ can be computed in time linear in $\text{Sl}^*(\pi)$.*

Proof. Consider a source–sink path P of $\text{Sl}^+(\pi)$. Let $\mathcal{S}$ be the set of minimal separation lines the vertices of P are labelled with. Due to Lemma 5.1, $\mathcal{S}$ is a maximal set of pairwise non-intersecting minimal separation lines of G . By Lemma 4.6 and the definition of the arc weights of $\text{Sl}^+(\pi)$, the largest arc weight on P is equal to the size of a largest clique of $G(\mathcal{S})$. Since there is a source–sink path of $\text{Sl}^+(\pi)$ for every minimal triangulation of G according to Corollary 4.8, the treewidth of G is equal to the shortest-path measure defined in statement 1. A source–sink path of $\text{Sl}^+(\pi)$ of smallest maximum arc weight can be computed in time linear in $\text{Sl}^+(\pi)$ by processing the vertices according to a topological ordering.

For the second statement, first consider an arbitrary interval graph H . The number of edges of H can be computed using only the cardinalities of the maximal cliques and their intersections. Let $\langle A_1, \dots, A_k \rangle$ be a consecutive clique arrangement for H . It holds that

$$|E(H)| = \binom{|A_1|}{2} + \sum_{i=2}^k \left(\binom{|A_i|}{2} - \binom{|A_{i-1} \cap A_i|}{2} \right) = \sum_{i=1}^k \binom{|A_i|}{2} - \sum_{i=2}^k \binom{|A_{i-1} \cap A_i|}{2};$$

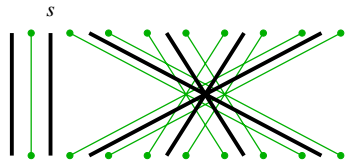


Fig. 2. Depicted is a permutation diagram, that is defined by the thin line segments. The thick line segments represent the minimal separation lines. Minimal separation line s has four successors.

this equivalence is straightforward to prove. Let P be a source–sink path of $SI^*(\pi)$ and let $\mathcal{S}$ be the set of minimal separation lines on P . By the result of Lemma 4.6 and arguments similar to the ones above, we see that the number of edges of $G(\mathcal{S})$ is equivalent to the arc-minus-vertex weight of P . It is important to note that the weights of source and sink vertex are 0, since $C(\mathbf{0}) = C(\mathbf{1}) = \emptyset$. Hence, the minimum fill-in of G is equal to the minimum number described in statement 2. And since a source–sink path of $SI^*(\pi)$ of smallest arc-minus-vertex weight can be computed in linear time by processing the vertices according to a topological ordering, we conclude the lemma. $\square$

5.2. Algorithms

The results in Section 5.1 show that all considered problems can be solved if the weighted minimal separation lines graph is given. We show in this subsection that the weighted minimal separation lines graph can be computed in linear time. This will lead to linear-time algorithms for the considered minimal triangulation problems. The computation of the weighted minimal separation lines graph is partitioned into three steps: (1) listing the minimal separation lines, (2) computing the arcs, (3) computing the weights. We consider the steps and give algorithms for them separately.

Lemma 5.4. *Let $G = G(\pi)$ be a permutation graph. The number of minimal separation lines of G is at most $1 + |V(G)| + |E(G)|$. The minimal separation lines of G can be listed in linear time.*

Proof. Let s be a minimal separation line of G with $\mathbf{0} < s \leq \mathbf{1}$. Let $\left\langle \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right\rangle$ be the enclosure of s . Note that a and c exist and are vertices of G . It holds that either $a = c$ or $ac \in E(G)$. Let t be a minimal separation line of G and let $\left\langle \begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix} \right\rangle$ be the enclosure of t . It holds that $s = t$ if and only if $a' = a$ and $c' = c$. Hence, every vertex and every edge of G defines at most one minimal separation line of G , and therefore, there are at most $1 + |V(G)| + |E(G)|$ minimal separation lines of G .

For listing the minimal separation lines, it suffices to check for every vertex x of G whether $(x + \frac{1}{2}, \pi^{-1}(x) + \frac{1}{2})$ is a minimal separation line of G and for every edge uv of G where $u < v$ whether $(v + \frac{1}{2}, \pi^{-1}(u) + \frac{1}{2})$ is a minimal separation line of G . This can be done in total linear time, since for a given scanline, computing the enclosure vertices and checking whether they are crossed by the scanline can be done in constant time. $\square$

The arcs of the weighted minimal separation lines graph are computed by listing the set of successors for every minimal separation line. Interestingly, the size of such a set has only a trivial upper bound, as we show with the example in Fig. 2. Minimal separation line s has four successors. It is not difficult to see that the example can be generalised to show that a minimal separation line can have an arbitrary number of successors. However, the many successors of a minimal separation line t cannot have an endpoint in common with t . This observation motivates that we consider the two types of successors, with or without common endpoint, separately. We begin with successors without common endpoint. For a permutation graph $G = G(\pi)$ and two scanlines s and t of G , we write $s \ll t$ if $s < t$ and s and t have no endpoint in common.

Lemma 5.5. *Let $G = G(\pi)$ be a permutation graph. Let s and t be minimal separation lines of G where t is a successor of s . Then, $R(s) \cap L(t) \neq \emptyset$ if and only if $s \ll t$.*

Proof. If s and t have a common endpoint then $R(s) \cap L(t) = \emptyset$. Let $s \ll t$. Suppose that $R(s) \cap L(t) = \emptyset$. Let $\left\langle \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right\rangle$ and $\left\langle \begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix} \right\rangle$ be the enclosure of respectively s and t . By assumption, $b \in C(t)$ and $c' \in C(s)$. Thus, b and c' are non-adjacent in G , and this contradicts the third statement of Lemma 3.5. Hence, $R(s) \cap L(t) \neq \emptyset$. $\square$

Similar to the algorithm for listing minimal separation lines, the main idea for listing successors is to establish a connection to objects that can be found and listed more easily. In our case, maximal clique-modules turn out useful. Let G be a graph. A set M of vertices of G is a *module* of G if every vertex of G that is not contained in M is adjacent to either every vertex in M or to no vertex in M . A module M of G is a *clique-module* of G if M is a clique of G . A clique-module M of G is called *maximal* if there is no clique-module of G that contains M as a proper subset. Clique-modules were first considered by Roberts in connection with his study of indifference graphs [15]. The set of maximal clique-modules of G is a partition of $V(G)$ [15]. For a permutation graph $G = G(\pi)$ and a clique-module C of G , we say that C is *compact* in G if the vertices in C appear consecutively in π .

Lemma 5.6. Let $G = G(\pi)$ be a permutation graph. A clique-module C of G is compact if and only if there are numbers a, k such that $C = \{a + i : 0 \leq i \leq k\}$.

Proof. Let C be a clique-module of G . If $|C| = 1$ then the lemma trivially holds. So, let $|C| \geq 2$. First, let C be compact. Suppose there are vertices $u, v \in C$ and $x \notin C$ such that $u < x < v$. Since C is compact, it holds $x \prec_\pi v \prec_\pi u$ or $v \prec_\pi u \prec_\pi x$, which means that x is non-adjacent to u or v . This is a contradiction to C being a module. The converse is proved analogously, or follows directly from flipping the permutation diagram along the horizontal line and renaming the vertices. $\square$

Lemma 5.7. Let $G = G(\pi)$ be a permutation graph. Let s and t be minimal separation lines of G where t is a successor of s and $s \ll t$. Let $\left\langle \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right\rangle$ and $\left\langle \begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix} \right\rangle$ be the enclosure of respectively s and t , and let $C =_{\text{def}} R(s) \cap L(t)$.

- (1) C is a compact maximal clique-module of G .
- (2) Let u be an arbitrary vertex from C . For every vertex x of G with $b \leq x \leq a'$ or $d \prec_\pi x \prec_\pi c'$, $x \in N_G[u]$.
- (3) $d, b \in C$ or $b, a' \in C$ or $a', c' \in C$ or $c', d \in C$.

Proof. Due to Lemma 5.5, C is not empty. Since all vertices of G outside of C that are adjacent to a vertex in C are contained in $C(s) \cup C(t)$, the results of Lemma 3.5 show that C is a clique-module of G . Suppose that C is not compact. Then, there are vertices $u, v \in C$ and a vertex $x \notin C$ with $u \prec_\pi x \prec_\pi v$. Since $x \notin C$, $x \in C(s) \cup C(t)$, which particularly means that x is non-adjacent to u or v . This is a contradiction to C being a module. Hence, C is a compact clique-module. Now, suppose that there is a vertex x such that $C \cup \{x\}$ is a clique-module of G . If $x \in C(s)$ then $0 < s$ and x is adjacent to a or c , if $x \in C(t)$ then $t < 1$ and x is adjacent to b' or d' . In both cases, x has neighbours that no vertex in C has, which yields a contradiction. Thus, C is a maximal clique-module.

For the second statement, note for a vertex x of G with $b \leq x \leq a'$ or $d \prec_\pi x \prec_\pi c'$ that $x \in C \cup C(s) \cup C(t)$. Due to Lemma 3.5, $x \in N_G[u]$.

For the third statement, assume that $b \notin C$. Since $b \in R(s)$, this means that $b \in C(t)$. If $c' \notin C$ then $c' \in C(s)$. Then, $\left\langle \begin{smallmatrix} a & b \\ c' & d' \end{smallmatrix} \right\rangle$ is the enclosure of a minimal separation line s' of G with $s < s' < t$. This yields a contradiction to t being successor s . Hence, $b \notin C$ implies $c' \in C$. With analogous arguments, we can show that $d \notin C$ implies $a' \in C$, which proves the statement. $\square$

Lemma 5.8. Let $G = G(\pi)$ be a permutation graph. The ordered pairs (s, t) of minimal separation lines of G where t is a successor of s and $s \ll t$ can be listed in linear time.

Proof. The algorithm relies on the properties described in Lemma 5.7:

step 1: list the compact maximal clique-modules of G

step 2: for every listed clique-module C , choose a vertex $u \in C$, determine the four (not necessarily distinct) vertices a, b, c, d where

- a is smallest with $a \leq u$ and for all x with $a \leq x < u$, $xu \in E(G)$
- b is largest with $u \leq b$ and for all x with $u < x \leq b$, $xu \in E(G)$
- $\pi^{-1}(c)$ is smallest with $c \prec_\pi u$ and for all x with $c \prec_\pi x \prec_\pi u$, $xu \in E(G)$
- $\pi^{-1}(d)$ is largest with $u \prec_\pi d$ and for all x with $u \prec_\pi x \prec_\pi d$, $xu \in E(G)$,

let $s_u =_{\text{def}} (a - \frac{1}{2}, \pi^{-1}(c) - \frac{1}{2})$ and $t_u =_{\text{def}} (b + \frac{1}{2}, \pi^{-1}(d) + \frac{1}{2})$, and

if $c, a \in C$ or $a, b \in C$ or $b, d \in C$ or $d, c \in C$ then add (s_u, t_u) to the solution set.

We show that this algorithm indeed lists the desired pairs of minimal separation lines. We first show that the defined scanlines are minimal separation lines, independent of the actual chosen vertex. Let C be a compact maximal clique-module of G and let u be the chosen vertex. Let a, b, c, d be the vertices that are computed for u according to step 2. Note that $a, b, c, d \in N_G[u]$ and $a \leq u \leq b$ and $c \prec_\pi u \prec_\pi d$. Thus, $a, c \notin C(s_u)$ and $b, d \notin C(t_u)$. Assume that $a \geq 2$ and consider vertex $a - 1$. If $a - 1 \in C(s_u)$ then $a - 1 \notin N_G(u)$ by the definition of a . Then, $c \prec_\pi a - 1 \prec_\pi u$, and this yields a contradiction to the definition of c . Thus, $a - 1 \notin C(s_u)$. By analogous arguments, it follows that no enclosure vertex of s_u is crossed by s_u and no enclosure vertex of t_u is crossed by t_u , so that s_u and t_u are minimal separation lines of G . Next, we show that s_u and t_u are unique for C , independent of the choice of u . Let $v \in C$ with $v \neq u$. Since C is a compact clique-module of G , $v \in R(s_u) \cap L(t_u)$, and no vertex from $L(s_u) \cup R(t_u)$ is adjacent to v . Let a', b', c', d' be the vertices that are defined for v according to step 2. Then, $a \leq a', b' \leq b, c \prec_\pi c'$ and $d' \prec_\pi d$. Suppose that $a < a'$, which means that there is a vertex x with $a \leq x < a'$ and $xv \notin E(G)$. This, however, is not possible by u and v belonging to the same clique-module. By similar arguments, we can show that $a = a', b = b', c = c'$ and $d = d'$. Hence, s_u and t_u are uniquely defined for C , i.e., independent of the actual choice of u . We have to check whether t_u is successor of s_u . Assume that there is a minimal separation line s' of G with $s_u < s' < t_u$. Let $\left\langle \begin{smallmatrix} a'' & b'' \\ c'' & d'' \end{smallmatrix} \right\rangle$ be the enclosure of s' . If s' does not cross u then a'' with $a < a'' < u$ or c'' with $c \prec_\pi c'' \prec_\pi u$ is not adjacent to u , or b'' with $u < b'' < b$ or d'' with $u \prec_\pi d'' \prec_\pi d$ is not adjacent to u . Both cases yield a contradiction, so that $u \in C(s')$. There are two cases to consider: $b'' < u$ or $u < a''$. Since both cases are symmetric, we only consider the former one; the latter one is proved analogously. Since $b'' < u$, it holds that $u \prec_\pi c''$. Since b'' and c'' are non-adjacent and neighbours of u , $b'' \notin C$ and $c'' \notin C$. Since C is compact and $a < b'' < u$ and $u \prec_\pi c'' \prec_\pi d$, $a \notin C$ and $d \notin C$, and (s_u, t_u) is not added to the solution set at the end of step 2. We conclude that if (s_u, t_u) is added to the solution set, t_u is successor of s_u .

For completing the correctness proof of the algorithm, it remains to show that the algorithm finds every pair. Let s and t be minimal separation lines of G where $s \ll t$ and t is successor of s . Let $\left\langle \begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix} \right\rangle$ and $\left\langle \begin{smallmatrix} a'' & b'' \\ c'' & d'' \end{smallmatrix} \right\rangle$ be the enclosure of respectively s and t . Let $C =_{\text{def}} R(s) \cap L(t)$. We apply Lemma 5.7 and obtain that C is a compact maximal clique-module of G , thus is listed in step 1 of our algorithm. Furthermore, vertices b', a'', d', c'' are exactly the vertices that are found as a, b, c, d in step 2, for any choice of u . Hence, $s = s_u$ and $t = t_u$ for s_u and t_u as in step 2. And by the result of statement 3 of Lemma 5.7, $(s, t) = (s_u, t_u)$ is added to the solution. We conclude that our algorithm correctly lists all desired pairs of minimal separation lines.

We consider running-time aspects. The partition of $V(G)$ into maximal clique-modules can be computed in linear time, for instance by partition refinement. Note that there are at most $|V(G)|$ maximal clique-modules of G . For computing vertices a, b, c, d in step 2, it suffices to visit only neighbours of the chosen vertex, so that the total computation time for all listed clique-modules is linear. The check for enclosure vertices belonging to clique-modules takes constant time per vertex, by a simple table look-up, and since there are at most $4|V(G)|$ vertices to check, we conclude total linear running time for the algorithm. $\square$

Theorem 5.9. *Let $G = G(\pi)$ be a permutation graph. The minimal separation lines graph of G , $Sl(\pi)$, can be constructed in linear time.*

Proof. The vertices of $Sl(\pi)$ correspond to the minimal separation lines of G , and the arcs of $Sl(\pi)$ are defined by the successor relation. The minimal separation lines can be listed in linear time due to Lemma 5.4. The arcs of $Sl(\pi)$ can be partitioned into the ones that correspond to pairs of minimal separation lines without a common endpoint and to pairs of minimal separation lines with a common endpoint. The former set of pairs can be listed in linear time due to Lemma 5.8. For the second type of pairs, we first observe that a minimal separation line can have at most two such successors. We define two orderings on the set of minimal separation lines. Let $\rightarrow$ be defined as follows: order the minimal separation lines increasingly by their upper endpoints and within an equivalence class by their lower endpoints. Formally, for separation lines $s = (p, q)$ and $t = (p', q')$, $s \rightarrow t$ if and only if $p < p'$ or if $p = p'$ then $q < q'$. Similarly, we define order $\rightarrow'$: $s \rightarrow' t$ if and only if $q < q'$ or if $q = q'$ then $p < p'$. Both orderings can be computed in linear time. Let s and t be minimal separation lines of G with $s < t$ where t is a successor of s and s and t have a common endpoint. This is the case if and only if s and t have the same upper endpoint, $s \rightarrow t$ and they appear consecutively in $\rightarrow$, or they have the same lower endpoint, $s \rightarrow' t$ and they appear consecutively in $\rightarrow'$. Given the two orderings, all such pairs can be listed in linear time by simply transversing the orderings. $\square$

Theorem 5.9 gives a linear-time algorithm for computing the minimal separation lines graph. We want to emphasise that $Sl(\pi)$ of $G = G(\pi)$ has at most $|V(G)| + 2(|V(G)| + |E(G)|)$ arcs. It remains to compute the weights. We first show how to compute the vertex weights efficiently.

Lemma 5.10. *Let $G = G(\pi)$ be a permutation graph. The vertex weights of $Sl^+(\pi)$ can be computed in linear time.*

Proof. Let G have n vertices. We compute for every minimal separation line s of G the cardinalities of $L(s)$ and $R(s)$, from which we will obtain $|C(s)|$. Consider the following auxiliary procedure, that iteratively combines elements in lists:

let $\mathcal{L}(n) =_{\text{def}} \langle (0, \{0\}), (1, \{1\}), \dots, (n, \{n\}) \rangle$;
 for every i from $n - 1$ down to 1 do:
 step A
 let (a, S) be the pair in $\mathcal{L}(i + 1)$ with $\pi(i + 1) \in S$;
 let (a', S') be the pair in $\mathcal{L}(i + 1)$ with $a' = a - 1$
 step B
 obtain $\mathcal{L}(i)$ by doing:
 $S' \leftarrow S \cup S'$, decrease b by 1 in every (b, T) where $b > a$, delete (a, S) .

In the following, we call the first component in the ordered pairs of the lists, the numbers a, a', b in the procedure, the *index* of the set in the second component. We first show some properties of the computed lists. For every $1 \leq i \leq n$:

- (1) the indices in $\mathcal{L}(i)$ are $0, 1, \dots, i$
- (2) for $i < n$, $a \geq 1$ for a as chosen in step A
- (3) the second component sets in $\mathcal{L}(i)$ define an ordered interval cover of the list $\langle 0, 1, \dots, n \rangle$
- (4) $\{0, \pi(1), \dots, \pi(i)\}$ is equal to the set consisting of the smallest element in each second component set in $\mathcal{L}(i)$.

It is directly clear that $\mathcal{L}(n)$ has the four properties. Consider $i < n$. Since $\pi(i) \geq 1$ is smallest element in a second component set in $\mathcal{L}(i + 1)$ and 0 is the smallest element in the ordered pair of $\mathcal{L}(i + 1)$ with index 0, it holds that $a \geq 1$. Consider the ordered pairs (a, S) and $(a - 1, S')$ in $\mathcal{L}(i + 1)$. By assumption, S and S' are consecutive intervals of $\langle 0, \dots, n \rangle$. Thus, $S \cup S'$ is an interval of $\langle 0, \dots, n \rangle$. The definition of $\mathcal{L}(i)$ in step B proves that $\mathcal{L}(i)$ has properties 1 and 3. For property 4, it remains to observe that by assumption, $\pi(i + 1)$ is the smallest element in S and no other element in S is among $\pi(1), \dots, \pi(i + 1)$, so that $S \cup S'$ contains exactly one element that is among $\pi(1), \dots, \pi(i)$ and this is the smallest element in $S \cup S'$.

We claim that for a minimal separation line $s = (p, q)$ of G , $|L(s)|$ is equal to the index of the pair in $\mathcal{L}(q - \frac{1}{2})$ whose second component set contains $p - \frac{1}{2}$. This can be seen as follows. Set $L(s)$ contains exactly the vertices among $\pi(1), \dots, \pi(q - \frac{1}{2})$ that are smaller than p , and the claimed equality follows from properties 1, 3, 4 of the algorithm.

For the running time, we first show that $\mathcal{L}(n), \dots, \mathcal{L}(1)$ can be computed in linear time. We do not save every $\mathcal{L}(i)$ but manipulate on the same structure. For every number $1, \dots, n$, a pointer to the containing list element is stored in an array. Every second component set is stored as a chained list. In a loop execution, the pair (a, S) with $\pi(i+1) \in S$ can be determined in constant time using the pointer array, and following a pointer, (a', S') is determined in constant time as well. Step B requires careful consideration. The union $S \cup S'$ is computed, which takes constant time by re-directing pointers. However, this union implies pointer changes in the array. Instead of changing the pointers to the elements in S , we change the pointers to the elements in S' . Note that all but one element in S' are neighbours of $\pi(i+1)$, so that changing the pointers takes time $\mathcal{O}(|N_G(\pi(i+1))|)$. Finally, for decreasing indices for pairs (b, T) with $b > a$, note that the smallest element in T is a neighbour of $\pi(i+1)$. Thus, decreasing these indices takes time $\mathcal{O}(|N_G(\pi(i+1))|)$. Therefore, we conclude amortised linear time for the auxiliary procedure. After every loop execution, the numbers for every minimal separation line $s = (p, q)$ with $q = i + \frac{1}{2}$ is determined. Since there are at most $1 + |V(G)| + |E(G)|$ minimal separation lines of G due to Lemma 5.4, $|L(s)|$ for each minimal separation line s of G can be computed in linear time.

With a symmetric procedure, $|R(s)|$ for each minimal separation line s of G can be computed in linear time. For a scanline s of G , it holds that $|C(s)| = n - |L(s)| - |R(s)|$. $\square$

Theorem 5.11. *Let $G = G(\pi)$ be a permutation graph. The weighted minimal separation lines graph of G , $Sl^+(\pi)$, can be constructed in linear time.*

Proof. Due to Theorem 5.9, $Sl(\pi)$ can be constructed in linear time. And due to Lemma 5.10, the vertex weights of $Sl^+(\pi)$ can be computed in linear time. It remains to compute the arc weights. Let (u, v) be an arc of $Sl^+(\pi)$. Let $s = (p, q)$ and $t = (p', q')$ be the minimal separation lines corresponding to respectively u and v . By definition, the weight of (u, v) is $|R(s) \cap L(t) \cup C(s) \cup C(t)|$. We claim that this number is equal to

$$\frac{1}{2} \cdot (|C(s)| + |C(t)| + (p' - p) + (q' - q)).$$

To see this equality, verify the following facts:

- every vertex in $C(s) \cap C(t)$ contributes exactly 2 to $|C(s)| + |C(t)|$
- every vertex in $R(s) \cap L(t)$ contributes exactly 2 to $(p' - p) + (q' - q)$
- every vertex in $(C(s) \cup C(t)) \setminus (C(s) \cap C(t))$ contributes exactly 1 to $|C(s)| + |C(t)|$ and to $(p' - p) + (q' - q)$
- other vertices do not contribute to the sum.

Hence, the arc weights of $Sl^+(\pi)$ can be computed by simply traversing the given partially weighted digraph. Since the number of vertices plus the number of arcs of $Sl(\pi)$ is bounded above by $c \cdot (|V(G)| + |E(G)|)$, we conclude the result of the theorem. $\square$

Corollary 5.12. *Let $G = G(\pi)$ be a permutation graph. The binomial-weighted minimal separation lines graph of G , $Sl^*(\pi)$, can be constructed in linear time.*

Proof. The weighted minimal separation lines graph of G can be constructed in linear time due to Theorem 5.11. For constructing the binomial-weighted minimal separation lines graph of G , it suffices to replace every weight a by $\binom{a}{2}$. This can be done in linear time. $\square$

The results in this subsection together show that the problems that are considered in Section 5.1 can be solved in linear time.

Theorem 5.13. *Let $G = G(\pi)$ be a permutation graph. The following can be computed in linear time:*

- the treewidth and minimum fill-in of G
- the number of minimal triangulations of G
- the smallest and largest number of maximal cliques in a minimal triangulation of G .

Proof. The results follow from Theorems 5.9 and 5.11, Corollary 5.12 and Lemmas 5.2 and 5.3. $\square$

Let $G = G(\pi)$ be a permutation graph on n vertices. Let $\mathcal{S}$ be a set of pairwise non-intersecting scanlines of G . It is clear that $|\mathcal{S}| \leq 2n + 1$. Based on π , the interval model $\mathcal{F}_G(\mathcal{S})$ for $G(\mathcal{S})$ can be computed in $\mathcal{O}(n)$ time. The problems in Theorem 5.13 are solved by finding source–sink paths in $Sl(\pi)$, i.e., by finding a maximal set of pairwise non-intersecting minimal separation lines that describes a minimal triangulation of G with the desired property. Thus, in linear time, an interval model of a minimal triangulation of G with, for example, smallest possible clique number can be computed.

Our approach to solving the considered problems also allows efficient solutions of further problems. For example, by deleting arcs of $Sl^+(\pi)$ that have weight more than $\text{tw}(G) + 1$, the number of minimal triangulations with clique number $\text{tw}(G) + 1$ can be computed in linear time. With a similar approach, a minimal triangulation of G with clique number $\text{tw}(G) + 1$ and smallest number of edges can be computed in linear time. In particular, it can be decided in linear time whether G has a minimal triangulation that minimizes the clique number and the number of edges at the same time.

6. Final remarks and conclusions

This paper presents as its main structural result a decomposition scheme for permutation graphs that allows to solve generally hard problems efficiently. This decomposition scheme, described as the minimal separation lines graph, partitions the graph into small well-structured pieces. We use the decomposition scheme for devising linear-time algorithms for computing the treewidth and minimum fill-in of permutation graphs.

Since minimal triangulations of permutation graphs are interval graphs, computing a minimal triangulation of a permutation graph (of certain properties) can be understood as a layout problem by the connection to interval orderings (see [12] for more details). There are well-studied and practically important layout problems whose complexity when restricted to permutation graphs is still unknown [8,7]. The most famous such problem is certainly the bandwidth problem. It seems possible that the tools that are developed in this paper, like the minimal separation lines graph, can help to solve these problems.

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